

MODULE-2

Biological treatment of the organic waste fraction; Direct land application, composting, and anaerobic digestion. MSW Rules 2016, Swachh Bharat Mission and Smart Cities Program.

Organic waste is biodegradable waste of plant or animal origin that can decompose in nature without too much difficulty and be transformed into another type of organic material.

The "**organic fraction of municipal solid waste**" (**OFMSW**) refers to the biodegradable component of household waste, like food scraps, coffee grounds, yard waste, and paper products, which can be broken down by microorganisms, making it suitable for biological treatment methods like composting or anaerobic digestion;

OFMSW and its biological treatment:

Composition: OFMSW includes food waste, plant material, paper, and other organic materials from households and businesses.

Importance: Separating OFMSW from other waste allows for its effective recycling through composting or anaerobic digestion, which can produce valuable biogas and compost.

Challenges: Contamination with non-biodegradable materials like plastic can hinder the composting process, so proper waste sorting is crucial.

Benefits of biological treatment:

Resource recovery: Converts waste into valuable products like compost (for soil amendment) or biogas (renewable energy source).

Reduced landfill waste: Diverting organic materials from landfills minimizes methane emissions.

The organic waste fraction of municipal solid waste should be treated because when left untreated, it significantly contributes to greenhouse gas emissions (particularly methane) when decomposing in landfills, contaminates soil and water, attracts pests, and wastes valuable nutrients that could be recycled as fertilizer; proper treatment allows for resource recovery through composting or anaerobic digestion, mitigating these environmental impacts and creating beneficial by-products like compost or renewable energy.

Why to treat organic waste?

- **Greenhouse gas reduction:** Organic waste decomposing in landfills produces methane, a potent greenhouse gas, which contributes significantly to climate change.
- **Environmental pollution:** Leachate from decomposing organic waste can pollute groundwater if not properly managed.
- **Odour and pest control:** Unmanaged organic waste attracts pests and generates unpleasant odours.
- **Resource recovery:** Organic waste can be composted to produce valuable soil amendment or used for anaerobic digestion to generate biogas, a renewable energy source.
- **Sustainability and circular economy:** Treating organic waste aligns with the principles of a circular economy by maximizing resource reuse and minimizing waste generation.

TREATMENT METHODS:

The following are the major treatment methods adopted for organic waste as shown in figure 1.

- **Direct Land Application:** Involves applying raw or partially treated organic waste directly onto the soil surface, where naturally occurring microorganisms break down the organic matter, converting it into nutrients that can be utilized by plants, essentially acting as a fertilizer while simultaneously improving soil health and structure;
- **Composting:** Aerobic decomposition of organic matter to produce nutrient-rich compost for soil amendment.
- **Anaerobic digestion:** Breakdown of organic material in the absence of oxygen to generate biogas (methane) for energy production.



Fig 1: Treatment of Organic Waste.

DIRECT LAND APPLICATION OF ORGANIC WASTE:

"Direct land application" of this organic fraction means directly applying the untreated organic waste to soil as a fertilizer, often used in agricultural settings, but with potential concerns regarding nutrient imbalances and contamination and shown in figure 2.. This method is commonly used in agriculture to improve soil fertility, enhance microbial activity, and recycle nutrients.

Steps for Direct Land Application

- Organic waste sources: Manure, food scraps, crop residues, sewage sludge, and green waste. Need to remove contaminants (plastics, metals, non-biodegradable materials).
- Spreading organic waste directly on top of the soil. Followed by mixing waste into the soil to enhance decomposition and minimize nutrient loss.
- For Liquid waste, directly injecting organic slurry into the soil to reduce odour and nutrient runoff.
- Soil microorganisms (bacteria, fungi, actinomycetes) break down organic matter and further, releases essential nutrients increasing soil microbial diversity and activity, improving overall fertility.
- Boosts natural biological processes, supporting plant growth and crop yield.

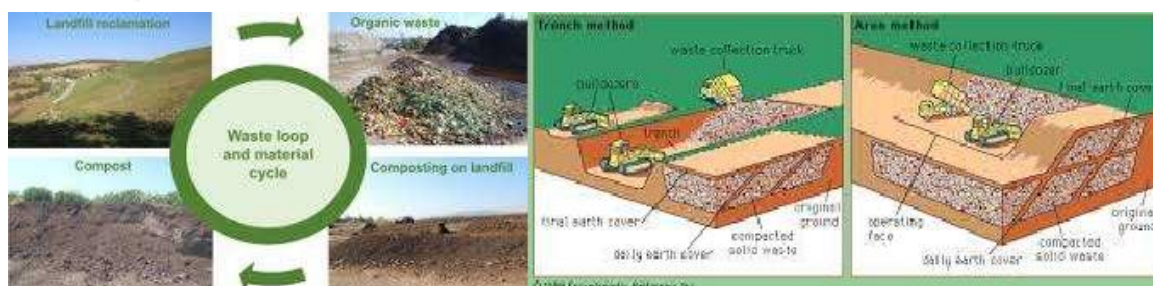


Fig 2: Direct Land Application

Factors affecting direct land application in waste management include: Waste composition, soil type, climate conditions, regulatory restrictions, land availability, potential contaminant levels in the waste, public perception, the presence of heavy metals or hazardous materials, application rates, and the intended use of the land; all of which must be considered to ensure safe and environmentally responsible land application practices.

- **Waste composition:** The type of waste being applied directly to land significantly impacts its suitability, with organic waste generally being more readily accepted than inorganic or hazardous materials.
- **Soil characteristics:** Soil type, texture, and drainage capacity determine how well the waste can be incorporated and the potential for nutrient leaching or contaminant migration.
- **Climate and weather:** Rainfall patterns and temperature can influence the decomposition rate of organic waste and the potential for nutrient runoff.
- **Regulations and legal frameworks:** Local regulations regarding land application practices, including permissible contaminant levels, must be adhered to.
- **Land availability:** Access to suitable land with appropriate soil conditions is critical for direct land application.
- **Contaminant levels:** Testing for heavy metals, pathogens, and other hazardous substances in the waste is essential to prevent environmental contamination.
- **Public perception:** Community acceptance of land application practices can be impacted by concerns about potential odors, aesthetics, and environmental risks.

Examples of specific factors depending on the type of waste:

- **Manure application:** Nutrient content, pathogen levels, application timing, and potential for ammonia volatilization.
- **Compost application:** Carbon-to-nitrogen ratio, maturity level, particle size, and potential for nutrient leaching.
- **Biosolids application:** Heavy metal content, pathogen levels, and potential for soil amendment benefits.

Advantages:

- Simple and cost-effective method, especially for small-scale operations.
- Provides nutrients to the soil, potentially improving soil health.
- Enhances soil water-holding capacity, reducing the need for irrigation.
- Carbon sequestration helps mitigate climate change by storing atmospheric CO₂ in the soil.
- Reduces landfill waste and mitigates environmental pollution.

Disadvantages:

- Potential for nutrient imbalances and excessive nitrogen loading if not properly managed.
- May introduce pests, pathogens, and weed seeds if not properly composted.
- Potential for odour issues and aesthetic concerns.

Important considerations for direct land application:

- **Waste quality:** Proper source separation is crucial to ensure the organic fraction is not contaminated with non-biodegradable materials. Use well-composted materials when possible to reduce pathogens.
- **Application rates:** Applying the right amount of organic waste to avoid nutrient excess and recommended to apply waste during dry weather to prevent runoff.
- **Soil analysis:** Monitoring soil nutrient levels to adjust application rates accordingly. Mix waste into the soil to minimize nutrient loss and odour.

COMPOSTING

Composting is a biological process that breaks down organic waste into a nutrient-rich fertilizer. It's a natural, economical, and eco-friendly way to treat organic waste. This process relies on **microorganisms** (bacteria, fungi, and actinomycetes) to break down organic matter under controlled conditions.

In municipal solid waste management, composting involves collecting the organic fraction of waste (like food scraps, yard waste) and allowing it to decompose naturally through the activity of microorganisms under controlled conditions, like proper moisture and aeration, to produce a usable compost material that can be used as fertilizer for soil improvement; this process typically involves steps like collection, sorting, pre-processing, mixing, turning the compost pile to ensure proper oxygenation, and monitoring the temperature to ensure optimal decomposition.

Steps in Composting:

- The first step is to separate the organic waste from other waste components like plastics and metals through sorting at the household or collection facility.
- The majority of composting methods used in municipal settings are aerobic, meaning they rely on oxygen-loving microbes to break down organic matter, producing carbon dioxide, water, and heat as by-products.
- The collected organic waste is typically placed in a compost pile or bin where the material is turned regularly to facilitate proper aeration and even decomposition as shown in figure 3.
- Maintaining a balanced carbon-to-nitrogen ratio is crucial for optimal composting; "brown" materials like leaves and wood chips provide carbon, while "green" materials like food scraps and grass clippings provide nitrogen.
- After the initial decomposition, the compost may go through a maturation phase where it is allowed to further stabilize and cure before being ready for use as a soil amendment.



Fig 3: Composting in Bins

Factors that affect composting include temperature, moisture, carbon-to-nitrogen ratio, aeration, and composition of waste.

- **Optimal temperature:** 131–160°F (55–71°C) is ideal for beneficial microorganisms
- **Temperature rise:** Microbial activity increases during compression, which causes a temperature rise
- **Pathogen destruction:** High temperatures kill pathogens that are harmful to plants, animals, and humans
- **Ideal moisture:** The pile should feel like a wrung-out sponge
- **Too dry:** Slows down the composting process
- **Too wet:** Creates an anaerobic environment that slows down decomposition and causes bad odors
- **High carbon-to-nitrogen ratio:** Limits the growth of bacteria and other microorganisms, slowing decomposition.
- **Forced aeration:** Air is passed through the pile continuously or periodically
- **Contained odours:** If air is drawn down the pile, odours from the compost are contained in the system
- **Texture of raw materials:** Affects the composting process
- **Volume of waste:** Affects the temperature

Advantages:

- Enhances soil structure, aeration, and moisture retention. The compost is a stable product that can be safely handled and stored.
- Provides slow-release nutrients (N, P, K) for plants. The compost can be used as a fertilizer or soil amendment.
- High temperatures kill harmful bacteria, parasites, and weed seeds.
- Reduces methane emissions compared to landfill disposal.
- Diverts organic waste from landfills and reduces overall waste volume.

Disadvantages (Challenges):

- Insufficient oxygen leads to anaerobic conditions and foul odours
- Too much or too little moisture slows down microbial activity.
- Full composting can take weeks to months, depending on conditions.
- Large-scale composting requires land and regular monitoring.

Types of Composting: There are several types of composting methods used to convert organic waste into nutrient-rich compost:



Backyard or Home Composting

This is a small-scale composting method suitable for households or small gardens. It involves composting kitchen scraps, yard waste, and other organic materials in a designated compost bin or pile.

Vermicomposting

Vermicomposting utilises earthworms to break down organic waste. Worms consume organic matter, and their castings (worm poop) create nutrient-rich compost. It can be done in small-scale setups like worm bins as shown in figure 4.



Fig 4: Vermicomposting

Aerobic Composting

Aerobic composting is a process of decomposing organic materials using oxygen-dependent microorganisms, such as bacteria and fungi, in the presence of air. This method typically involves creating a compost pile or using a compost bin with adequate airflow as shown in figure 5. The microorganisms break down the organic waste, generating heat as a by-product, which speeds up the decomposition process. This method helps to control odours, suppress pathogens and weed seeds, and maintain a balanced carbon-to-nitrogen (C: N) ratio.



Fig 5: Aerobic Composting



Fig 6: Anaerobic Composting

Anaerobic Composting

Anaerobic composting is a process of organic waste decomposition that occurs in the absence of oxygen. Unlike aerobic composting, which relies on the activity of oxygen-dependent microorganisms, anaerobic composting involves the breakdown of organic matter by anaerobic bacteria in an oxygen-deprived environment. As shown in figure 6, this process typically occurs in sealed containers or anaerobic digesters. Anaerobic composting tends to be slower and less efficient than aerobic composting, and it can produce odours and methane gas, a potent greenhouse gas.

Windrow Composting

Windrow composting involves creating long, narrow piles or rows of organic waste. These rows are periodically turned and aerated to promote decomposition. Windrow composting is often used in large-scale operations as shown in figure 7.



Fig 7: Windrow Composting



Fig 8: In-Vessel Composting

In-vessel Composting

In-vessel composting is done in enclosed containers or structures, such as compost tumblers or specially-designed composting machines as shown if figure 8. The controlled environment allows for faster and more efficient decomposition.

Aerated Static Pile Composting

In aerated static pile composting, organic waste is piled into a heap and then aerated using blowers or pipes to maintain proper oxygen levels as shown in figure 9. This method ensures efficient decomposition and odour control.

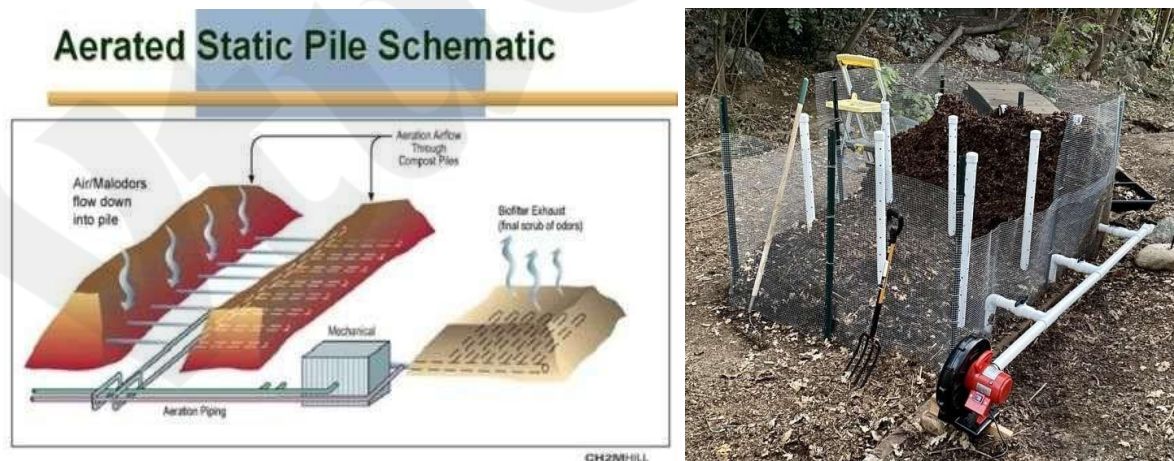


Fig 9: Aerated Static Pile Composting

Bokashi Composting

Bokashi composting is a fermentation process that involves adding a specialised mixture of beneficial microorganisms to organic waste as shown in figure 10. The waste is fermented in an airtight container, producing a nutrient-rich compost.

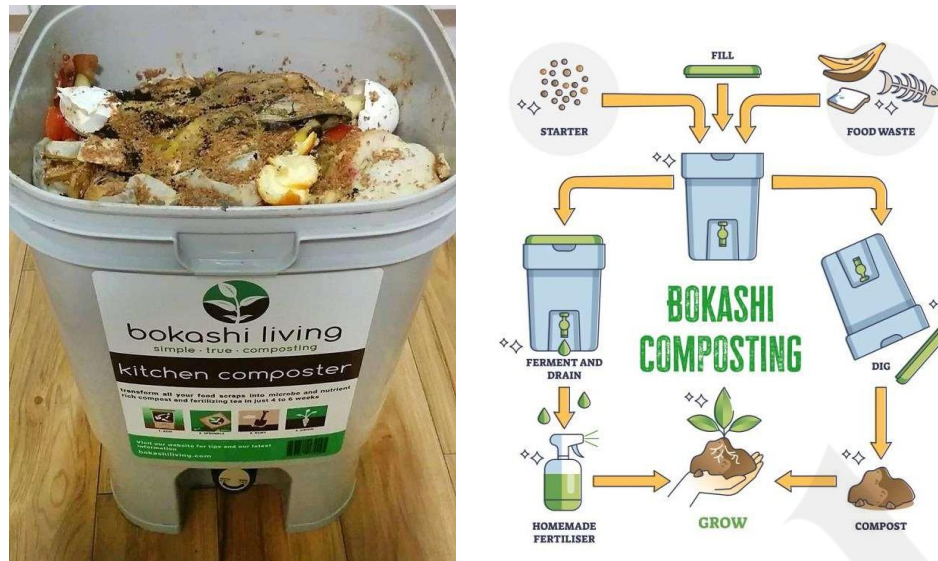


Fig 10: Bokashi Composting

Comparison: Direct Land Application vs. Composting:

Aspect	Direct Land Application	Composting
Decomposition Site	In soil (natural conditions)	In controlled compost piles
Processing Time	Immediate application, slow breakdown	4–12 weeks (faster breakdown)
Pathogen Risk	Higher (raw waste)	Lower (high heat kills pathogens)
Odour Control	Potential odour issues	Better odour management
Nutrient Release	Fast but variable release	Slow, steady nutrient supply

ANAEROBIC DIGESTION

Anaerobic digestion is a biological process that treats organic waste by breaking it down into biogas and other organic compounds. It's a common method for treating waste in both developed and developing countries.

Aerobic digestion in municipal solid waste treatment involves exposing the organic fraction of the waste to oxygen-rich conditions, allowing aerobic bacteria to break down the organic matter into carbon dioxide, water, and a stable compost-like material, essentially mimicking a natural composting process where microorganisms actively consume the organic material in the presence of oxygen; this method is often used to treat food scraps, yard waste, and other biodegradable components of municipal solid waste.

The figure 11 shows the working of anaerobic digestion.

1. **Pre-treatment:** The waste is homogenized and any contamination is removed.
2. **Anaerobic reactor:** The waste is placed in an airtight reactor without oxygen.
3. **Anaerobic digestion:** Microorganisms break down the waste into biogas and other organic compounds.
4. **Biogas capture:** The biogas is cleaned and captured in a storage balloon.
5. **Biogas use:** The biogas can be used to generate heat, electricity, or vehicle fuel.
6. **Sludge treatment:** The remaining sludge can be used as fertilizer, animal bedding, or for crop irrigation.

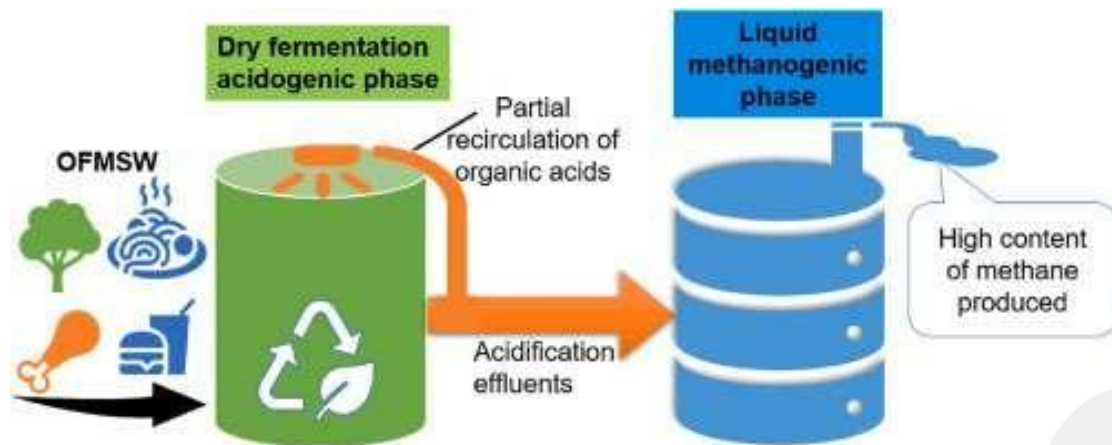


Fig 11: Working of Anaerobic Digestion

The process of biological Decomposition of Anaerobic Digestion is shown in figure 12.

1. Waste Collection & Pre-Treatment

- Organic waste sources: food waste, manure, sewage sludge, crop residues, and industrial byproducts.
- Sorting & pre-treatment: Remove contaminants (plastics, metals) and adjust moisture & particle size.

2. Biological Decomposition in an Anaerobic Reactor (*Four Key Stages*)

- a) Hydrolysis:** Complex organic matter (carbohydrates, proteins, fats) is broken down into simpler molecules (sugars, amino acids, fatty acids).
- b) Acidogenesis:** Microorganisms convert these molecules into organic acids, alcohols, and hydrogen.
- c) Acetogenesis:** Organic acids are further converted into **acetate, hydrogen, and carbon dioxide**.
- d) Methanogenesis:** Methanogenic bacteria produce **methane (CH₄) and carbon dioxide (CO₂)** from acetate and hydrogen.

3. End Products of Anaerobic Digestion

- **Biogas:** A renewable energy source (50-70% methane, 30-50% CO₂) used for electricity, heat, or fuel.
- **Digestate:** A nutrient-rich residue that can be used as an organic fertilizer for soil improvement.

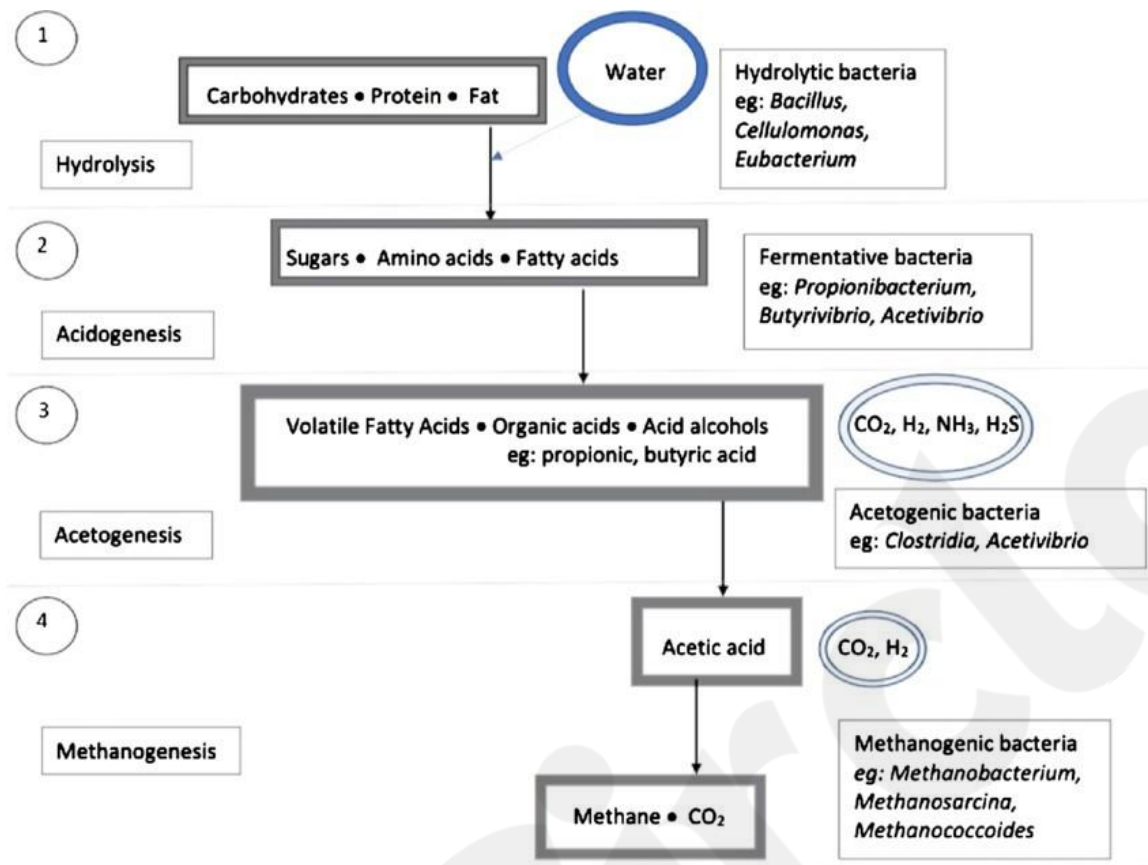


Fig 12: Biological Decomposition of Anaerobic Digestion

Factors affecting anaerobic digestion included are C/N ratio, F/M ratio, pH, temperature, Organic Loading Rate, Hydraulic Loading Rate, presence of toxins (Inherent component, or a by-product of the metabolism).

Benefits of Anaerobic Digestion as a Biological Treatment

- **Renewable Energy Production:** Converts organic waste into **biogas**, reducing fossil fuel dependence.
- **Organic Waste Management:** Reduces the volume of waste material, diverts waste from landfills and thus reducing methane emissions.
- **Nutrient Recycling:** Digestate is a valuable fertilizer that enhances soil health.
- **Odour & Pathogen Reduction:** Controlled conditions significantly reduce unpleasant odours and kill harmful pathogens.
- **Carbon Footprint Reduction:** Captures methane emissions that would otherwise contribute to climate change.

Challenges & Considerations

- **High Initial Investment:** Setting up an anaerobic digestion plant requires costly infrastructure.
- **Complex Operation & Maintenance:** Requires technical expertise for proper management. The quality of the end products determines whether they can be recycled or disposed of.
- **Process Stability:** Sensitive to temperature, pH, and feedstock composition.
- **Nutrient Imbalance:** Excess nitrogen or acidic conditions can disrupt microbial activity.

- **Biogas Purification Needed:** Raw biogas contains impurities (H_2S , moisture) that require **cleaning** before use.

Comparison: Anaerobic Digestion vs. Composting

Aspect	Anaerobic Digestion (AD)	Composting
Oxygen Requirement	Anaerobic (No oxygen)	Aerobic (Needs oxygen)
Main Product	Biogas (energy) + Digestate (fertilizer)	Compost (fertilizer only)
Pathogen Reduction	High (biogas process kills pathogens)	Moderate (requires temps)
Odour Control	Better controlled	Opposable odours if not Managed well
Energy Production	Yes (biogas)	No energy output
Nutrient Recycling	Yes (digestate applied to soil)	Yes (compost applied to soil)

MUNICIPAL SOLID WASTE RULES

Features of MSW Rules, 2016 (as per CPCB Notes)

1. *Applicability:*

- Applies to all generators, including residential, commercial, industrial, and institutional establishments.
- Covers urban and rural areas, special economic zones (SEZs), airports, railway stations, and places of worship.

2. *Waste Segregation at Source:*

- Waste must be segregated into three categories:
Biodegradable Waste (wet waste)
Non-Biodegradable Waste (dry waste)
Domestic Hazardous Waste (batteries, electronic waste, etc.)
- Segregation is the responsibility of households, offices, and all other waste generators.

3. *Collection and Transportation:*

- Local bodies must ensure door-to-door collection.
- Transportation must be in covered vehicles to prevent spillage and littering.

4. *Processing and Disposal:*

- Waste must be treated using appropriate technologies:
 - Composting, vermicomposting, anaerobic digestion, or bio-methanation for organic waste.
 - Material recovery facilities (MRFs) for recyclables.
 - Waste-to-Energy (WTE) plants for non-recyclables.
 - Sanitary landfilling only for residual and inert waste.
- Open dumping is strictly prohibited.

5. *Role of Waste Generators:*

- Households and institutions must segregate waste at the source.
- Bulk waste generators (such as large hotels, offices, and institutions) must process their waste.
- Plastic waste must be managed as per Plastic Waste Management Rules, 2016.

6. *Extended Producer Responsibility (EPR):*

- Manufacturers and brand owners must take responsibility for collecting and recycling their product packaging waste (e.g., plastic, multi-layered packaging).

7. *Role of Urban Local Bodies (ULBs):*

- Ensure scientific waste management, including proper treatment and disposal.
- Impose penalties for non-compliance.
- Create awareness campaigns.

8. *Prohibition of Open Burning of Waste:*

- Burning of solid waste is banned, and violators are subject to fines.

9. *Promotion of Decentralized Waste Processing:*

- Cities are encouraged to set up decentralized composting and biogas plants.

10. *Time-Bound Compliance Targets:*

- Deadlines set for phased implementation of waste processing facilities and landfills.

Penalties for Non-Compliance:

- Non-compliance with MSW Rules, 2016, can lead to fines imposed by local authorities.



Fig 13: Highlights of MSW Rules, 2016

The **MSW Rules, 2016** aim to promote **sustainable waste management** by enforcing segregation, processing, and disposal of municipal solid waste in an eco-friendly manner as shown in figure 13. The rules emphasize **scientific waste management** and the **shared responsibility** of citizens, businesses, and local authorities.

The **Ministry of Environment, Forest and Climate Change (MoEF)** will monitor the implementation of **Solid Waste Management** Rules in India. Besides, the ministry will create a Central Monitoring Committee under the Chairmanship of the Secretary, MoEF.

Areas of Applications of Solid Waste Management Rules in India

The rules shall apply to the following areas –

- Outgrowths in urban agglomerations,
- Every urban local body
- Notified areas
- Census towns as stated by the Census Commissioner and Registrar General of India
- Notified industrial townships
- Airports
- Areas under the control of Indian Railways
- Airbases
- Ports and harbours

- Special economic zones
- Defence establishments
- Pilgrim places of historical and religious importance as may be notified by the respective State governments from time to time
- State and Central government organisations
- Every institutional, domestic, commercial and any other non-residential solid waste generator

Duties of various authorities and personnel involved in effective Solid Waste Management Rules, 2016 implementation

1. Duties of waste generators

By definition, as stated in the **Solid Waste Management Rules, 2016**, waste generator means every individual or group of persons, every residential and non-residential premise such as defence establishments and Indian Railways. Every waste generator must perform the following duties to efficiently execute these rules.

- (a) Segregate and store the waste generated in three categories: non-biodegradable, bio-degradable and domestic hazardous wastes in specific bins. Subsequently, this segregated waste must be handover to authorised waste collectors or waste pickers.
- (b) Securely wrap the used sanitary waste and place them in the non- bio-degradable or dry waste bin.
- (c) Separately store and dispose of demolition, construction, horticulture and garden waste.
- (d) No waste generator shall burn, throw, or bury solid waste in open public spaces, streets, water bodies, or drain.
- (e) All waste generators must pay a user fee for solid waste management. As per the definition in **Solid Management Rules, 2016**, a user fee is a fee imposed on the waste generator by the local body and any entity indicated in rule 2 to offer waste management facilities.
- (f) Organising an event with more than one hundred people at a place without licence without informing the local body isn't allowed. Applicant must obtain approval by informing at least three working days in advance.
- (g) Every street vendor must have suitable containers for storage of waste generated and shall submit the waste at a waste storage depot, vehicle, or container as mentioned by the local body.
- (h) All resident welfare and market associations or gated institutions and communities having more than 5,000 sqm area or hotels and restaurants in partnership with the local body shall ensure effective management of the generated waste within an year from the date of publication of SWM rules.

2. Duties of the Ministry of Environment, Forest and Climate Change (MoEF)

Besides monitoring the implementation of **Solid Waste Management Rules, 2016** in India, the ministry will set up a Central Monitoring Committee under the Chairmanship of Secretary, MoEF, which will meet at least once yearly to review and monitor the implementation of these rules. The Committee must be renewed every three years.

3. Duties of Ministry of Agriculture, Government of India (GoI)

The Ministry of Agriculture, through appropriate mechanisms, must –

- (a) Offer flexibility in Fertiliser Control Order for selling and manufacturing compost.
- (b) Publicise usage of compost on farmland.
- (c) Establish laboratories to test compost quality produced by local authorities or their authorised agencies.

(d) Issue guidelines for maintaining the ratio of use and quality of compost and chemical fertilisers while using compost on farmland.

4. Duties of the Central Pollution Control Board

The Central Pollution Control Board must –

- (a) Co-ordinate with the Pollution Control Committees (PCC) and the State Pollution Control Boards (SPCB) for implementing **Solid Waste Management Rules** and adhering to the prescribed standards set up by local authorities.
- (b) Establishing the regulations for ambient air, groundwater, noise pollution, and leachate regarding all disposal facilities and solid waste processing.
- (c) Review environmental norms and standards mentioned for solid waste processing units or treatment technologies and improve them when required.
- (d) Review the implementation of prescribed environmental standards at least once a year and compile the data monitored.
- (e) Review the proposals of PCC or SPCB on using new technologies for recycling, processing and treatment of solid waste. Additionally, prescribe emission norms and performance standards within six months.
- (f) Monitor the implementation of these rules by local bodies.
- (g) Publish guidelines for maintaining buffer zone and on environmental aspects of disposal and processing of solid waste to allow local bodies to adhere to the provisions of these rules.
- (h) Guide Union Territories or States on inter-state movement of waste.

5. Duties of Pollution Control Committee (PCC) or State Pollution Control Board (SPCB)

The Pollution Control Committee or State Pollution Control Board must-

- (a) Enforce **Solid Waste Management Rules, 2016**, in their state and review implementation of these rules at least two times in a year.
- (b) Monitor environmental standards and adhere to Schedule I and II conditions for waste processing and disposal sites.
- (c) Analyse the authorisation proposal and make such inquiries as considered fit after receiving an application in Form I from the LB or any other authorised agency.
- (d) Grant authorisation within two months in Form II to the local body.
- (e) Synchronise the authorisation validity with the validity of the consent.
- (f) In case of new technologies, where PCC, CPCB or SPCB has not prescribed any standards, shall approach CPCB to get standards specified.
- (g) PCC or SPCB may instruct local bodies on the safe disposal and handling of domestic hazardous waste collected by the waste generators at hazardous waste deposition units.
- (h) PCC or SPCB will regulate the Inter-State movement of waste.

The process to get authorisation under Solid Waste Management Rules, 2016

To get the authorisation, every local authority of census towns and village panchayat must apply Form 1 to establish a waste processing, treatment, or disposal facility. If waste exceeds five metric tonnes per day, then the following documents are required: –

- Documents showing land ownership for solid waste processing site
- Project report on Solid Waste Management
- Municipal Solid Waste Annual Report of the previous year
- Work order or authorisation issued to the operator of the solid waste processing facility
- A permit from district level site selection committee

- Technical details of waste to energy plant or recycling plant
- Once the complete application and supporting documents are submitted, SPCB examines the proposal for authorisation
- After that, SPCB must issue authorisation within two months under Form 2 to village panchayat and local authorities running waste treatment, processing, or disposal facility.
- In case of cancellation of authorisation due to the facility failing to fulfil the conditions specified under the rules, the local body or operator should be served with proper notice.

When can the authorisation get cancelled or suspended?

According to new **Solid Waste Management Rules**, PCC or SPCB can suspend or cancel the authorisation issued under clause (a) at any time if the local operator or body of the facility fails to run the facility according to the norms specified: provided that no such authorisation shall be cancelled or suspended without giving notice to the local operator or body.

Renewal of the authorisation

On receiving the renewal application, PCC or SPCB renew the authorisation for the next five years. This is done after scrutinising every application on merit and according to the condition that the facility's operator has fulfilled all the provisions of the standards, rules, or conditions mentioned in the consents, authorisation, or environment clearance.

However, PCC or SPCB shall refuse to grant or renew an authorisation after offering enough chances to the applicant of being heard and for reasons thereof to be recorded in writing.

6. Duty of brand owners or manufacturers of disposable products and sanitary napkins and diapers

- (a) All manufacturers of disposable products like plastic packaging, glass and so on or brand owners introducing such products in the market must offer required financial assistance to local authorities for setting up of solid waste management system.
- (b) All such brand owners who market or sell their products in non-biodegradable packaging material must establish a system to gather back the packaging waste generated due to their production.
- (c) Brand owners, manufacturers, or marketing companies of diapers and sanitary napkins must explore the odds of utilising all recyclable materials in their products. Alternatively, they shall offer a wrapper or pouch for disposal of each diaper or napkin along with the packet of their sanitary products.
- (d) All such brand owners, manufacturers or marketing companies must be aware of the masses for wrapping and disposal of their products.

Criteria mentioned in Solid Waste Management Rules for duties regarding establishing solid waste processing and treatment facility

- (1) The concerned department of the allocation of the land assignment will be held responsible for offering suitable land for the establishment of the solid waste treatment and processing facilities and notify such sites by the Union Territory Administration or State Government.
- (2) The facility's operator must design and establish the facility according to the technical guidelines issued by the CPCB regarding this at regular intervals and the manual on SWM prepared by the Ministry of Urban Development.
- (3) The facility's operator must get necessary approvals from the PCC or SPCB.
- (4) The PCC or SPCB must analyse the environmental standards of the operation of these facilities.
- (5) The operator of the facility will be responsible for the secure and environment-friendly operations of the SW processing and/or treatment until according to the norms set up by the CPCB at regular intervals and the manual on Municipal SWM issued by the Ministry of Urban Development and updated regularly.

(6) The solid waste processing and treatment plant operator must submit an annual report in Form III every year by 30th April to the PCC or SPCB and the concerned local body.

Time Frame for Implementation of Solid Waste Management Rules:

- (a) Identifying landfill: 1 year
- (b) Ensure segregation of waste: 2 years
- (c) Cities up to 1 million population: 2 Years
- (d) Procurement of waste processing facilities: 2 years
- (e) Setting up sanitary landfills: 3 years
- (f) Bioremediation/capping of old landfills: 5 years
- (g) Million plus cities: 3 years

An essential revision of parameters and existing standards made under Solid Waste Management Rules, 2016

Under **Solid Waste Management Rules, 2016**, essential updates clearly state that the deposit site should be 200 meters from a pond, 100 meters from the river, 200 meters from habitations, highways, water supply wells and public parks and 20 km away from airbase or airports.

Updates are made on the emission standards, including furan, dioxin and particulate matter reduction limits.

The compost standards were also updated to comply with the Fertiliser Control Order.

Conclusion

It will take some years to observe the drastic change that **Solid Waste Management Rules 2016** will bring to India. Seeing how segregation at source works on the ground is challenging and exciting. Awareness campaigns with NGOs, communities, students and other stakeholders may work wonders to push for better implementation of SWM Rules. Moreover, the rules need to focus on awakening people about concerns, issues and management of solid waste.

SWACHH BHARAT MISSION was initiated by Prime Minister Narendra Modi to spread awareness and cleanliness across the country and was launched on October 2, 2014.

Objectives of Swachh Bharat Mission

- Hygiene and cleanliness should be promoted comprehensively.
- To reduce the number of people and households who defecate in the open.
- Raising the standard of living in rural communities.
- Encourage the use of environmentally appropriate sanitation.
- To raise public awareness about health and hygiene concerns
- To assist India in realising the aim of sustainable development.
- To encourage the use of cost-effective and efficient sanitation solutions.
- To concentrate on scientific waste management strategies for both solid and liquid waste.
- To have a positive impact on social inclusion and gender equality.

Mission Approach:

- Elimination of open defecation
- Eradication of Manual Scavenging
- Modern and Scientific Municipal Solid Waste Management
- To effect behavioural change regarding healthy sanitation practices

- Generate awareness about sanitation and its linkage with public health
- Capacity Augmentation for ULB's
- To create an enabling environment for private sector participation in Capex

Components:

- Household toilets, including conversion of insanitary latrines into pour-flush latrines;
- Community toilets
- Public toilets
- Solid waste management
- IEC & Public Awareness
- Capacity building and Administrative & Office Expenses (A&OE)

Strategy

- Comprehensive Sanitation Planning, which includes
 - (a) City Level Sanitation Plans
 - (b) State Sanitation Concept As per Annexure IV
 - (c) State Sanitation Strategy
- Behavioural Change Strategy and IEC
- Enabling Environment for Private sector participation
- Capacity Building

Special focus group

- Manual scavengers in urban areas (Employment in Sanitary toilets).
- Integration of Informal sector workers in waste management (rag pickers)
- Temporary accommodation for migrants and the homeless in urban areas
- Adequate provision for toilets either on the premises or linked to a public / community toilet.
- Mandating that construction labour in urban areas have access to toilets at all sites.
- Priority shall be accorded pro-actively to cover households with vulnerable sections such as pensioners, girl children, pregnant and lactating mothers.

Mission Management Structure:

- National Advisory and Review Committee (National Level, Headed by Secretary, MoUD)
- SBM National Mission Directorate (Headed by National Mission Director)
- High Powered Committee (State level, Headed by State's Chief Secretary)
- SBM State Mission Directorate
- District Level Review and Monitoring Committee (DLRMC)
- Urban Local Body level

SMART CITIES PROGRAM

The objective of SCM is to promote cities that provide core infrastructure and give a decent quality of life to its citizens, a clean and sustainable environment through the application of 'Smart' solutions and the components are shown in figure 12.

Smart City Mission

- Smart Cities Mission is to promote cities that provide core infrastructure and give a decent quality of life to its citizens, a clean and sustainable environment and application of 'Smart' Solutions.

- The Union Ministry of Urban Development is responsible for implementing the mission in collaboration with the state governments.
- The government of India has a vision of developing 100 smart cities by modernizing the existing mid-sized cities.



Fig 15: Smart City Mission

Core Infra structure Elements.

- Adequate water supply.
- Assured electricity supply.
- Sanitation, including solid waste management.
- Efficient urban mobility and public transport.
- Affordable housing, especially for the poor.
- Robust IT connectivity and digitalization.
- Good governance, especially e-Governance and citizen participation.

- Sustainable Environment.
- Safety and security of citizens, particularly women, children and the elderly.
- Health and Education.

SMART CITY WASTE MANAGEMENT

“Enhancing neighbourhoods, reducing costs with smart city waste management.”

As cities work to cost-efficiently manage waste in public spaces, smart city waste management programs can help to improve efficiency, reduce costs and enhance the beauty of public areas.

Components of Integrated Waste Management in a Smart City as shown in figure 16:

- **Source Segregation:**

Encouraging citizens to separate waste at home into categories like dry waste (plastics, paper), wet waste (food scraps), and hazardous materials, enabling efficient recycling and treatment.

- **Smart Bins with Sensors:**

Using IoT-enabled waste bins that monitor fill levels, providing real-time data to optimize collection routes and reduce unnecessary pickups.

- **Waste Tracking System:**

Monitoring the movement of waste from collection points to processing facilities through digital platforms, ensuring accountability and transparency.

- **Optimized Collection Routes:**

Utilizing data analytics to plan efficient collection routes based on waste generation patterns, minimizing fuel consumption and vehicle emissions.

- **Community Engagement:**

Educating citizens about proper waste disposal practices through mobile apps, awareness campaigns, and incentives for responsible waste management.

- **Waste-to-Energy Initiatives:**

Exploring technologies to convert organic waste into energy sources like biogas, contributing to renewable energy generation.

Benefits of Integrated Waste Management in Smart Cities:

- **Reduced Environmental Impact:**

Minimizing landfill waste, reducing greenhouse gas emissions, and conserving natural resources.

- **Cost Efficiency:**

Optimizing waste collection operations, reducing transportation costs, and maximizing recycling revenue.

- **Improved Public Health:**

Reducing potential disease vectors by managing waste effectively.

- **Citizen Empowerment:**

Fostering a sense of community responsibility towards waste management by providing accessible information and feedback mechanisms.

Examples of Smart Waste Management Technologies:

- **Waste App:**

Mobile applications for citizens to report waste collection issues, schedule pickups, and access recycling information.

- **Waste Compactor Monitoring:**

Sensors on waste compactors to track fill levels and optimize collection schedules.

- **Drone Surveillance:**

Aerial monitoring of waste collection areas to identify illegal dumping and potential environmental hazards.



Fig 14: Components of Integrated Waste Management in a Smart City

The benefits of a smart waste management system include:

- **Lower costs.** By having crews only make trips to bins that need attention, rather than visiting every bin on a regular basis, cities can reduce the cost of labor and fuel.
- **Less wear and tear.** Fewer trips also allows cities to extend the life of waste management vehicles and other assets.
- **Improved cleanliness in public areas.** Smart city waste management technology allows crews to empty bins before they become overflowing with trash or recycling, and before infestation becomes an issue. Smart waste sensors can also alert crews when bins develop unpleasant smells which can then be treated to eliminate odors.
- **Reduced congestion and carbon emissions.** Trash and recycling collection vehicles typically make many stops during a route, adding to congestion problems on city streets. By enabling crews to make fewer trips, smart waste management technologies help minimize congestion and reduce the pollutants emitted by trash and recycling collection vehicles.

More effective management. With detailed data about the status of bins throughout the city, public works departments can more easily manage the many vendors who are typically employed to handle different kinds of trash and recycling collection throughout the city